



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Earthlings

Team Leader Name : Shreyash Chaurasia

Problem Statement : Infrared Image Colorization and Enhancement for Improved Object Interpretation

Team Members

Team Leader:

Name: Shreyash Chaudasia
College: IIIT Vadodara

Team Member-1:

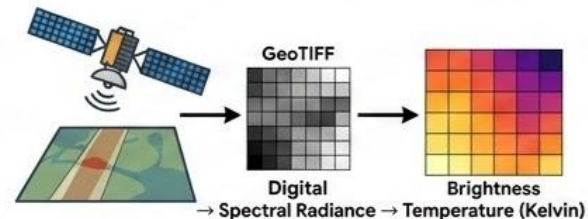
Name: Kavya Siddh Sharma
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Team Member-2:

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Opportunity should be able to explain the following:

- How different is it from any other existing ideas?



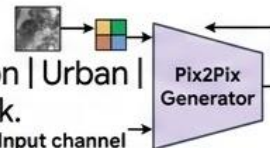
Most existing IR colorization approaches treat thermal imagery like a grayscale photograph, feeding normalized PNGs into a standard image-to-image model. We reject this approach entirely.

Our pipeline is built on three key differentiators:

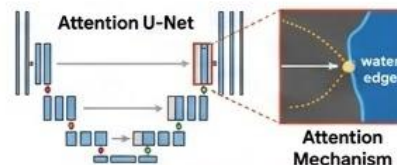


• **Physics-first input:** We work on raw Landsat 9 Band 10 GeoTIFF sensor values, converting Digital Numbers → Spectral Radiance → Brightness Temperature (Kelvin) before any model sees the data. No other team is likely doing this.

• **Semantic conditioning:** A pretrained SegFormer generates per-pixel land-cover masks (Water | Vegetation | Urban | Soil) that are concatenated as input channels into the colorization model, not just used as a post-hoc check.



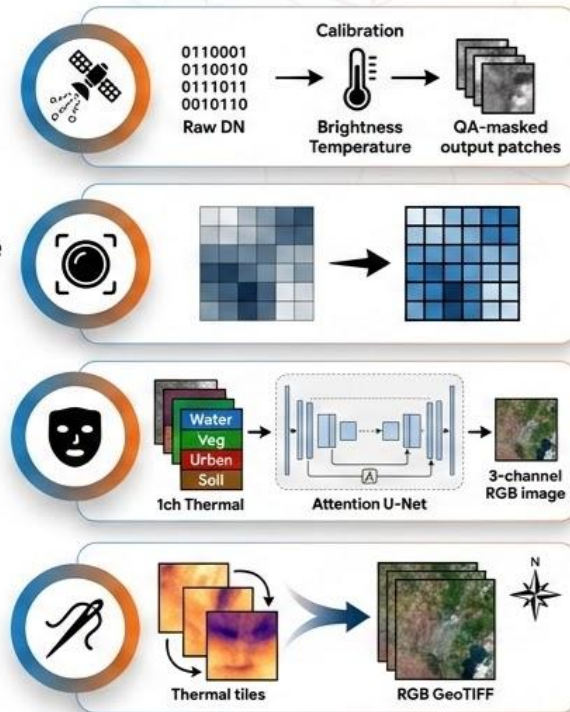
• **Attention mechanism:** Our generator uses an Attention U-Net instead of a vanilla U-Net, learning to focus on semantically critical boundaries (water edges, road margins) that are easily lost in low-contrast thermal data.



How it will be able to solve the problem ?

The pipeline operates in 4 sequential stages:

- 1. Physics-aware preprocessing:** converts raw sensor values to brightness temperature, applies QA-band cloud masking, and extracts spatially consistent 256×256 patches.
- 2. Super-Resolution (Real-ESRGAN):** sharpens faint edges and textures in the thermal image before colorization, recovering 30m- equivalent detail from 100m Band 10 data.
- 3. Semantic-guided colorization (Pix2Pix + Attention U-Net):** takes a 5-channel input (1ch SR thermal + 4ch land-cover mask) and predicts a 3-channel RGB image using a composite loss: Adversarial + L1 + VGG Perceptual + Semantic Constraint.
- 4. Post-processing:** applies histogram matching to Landsat Bands 4-3-2 reference and overlap-weighted tile stitching to a seamless, geospatially correct full-scene RGB GeoTIFF.



Output is evaluated against ground-truth Bands 4-3-2 using PSNR, SSIM, LPIPS, and FID.

USP of the proposed solution

The only pipeline that combines physics-aware radiometric preprocessing + super-resolution sharpening + semantic-conditioned colorization into a single, scientifically rigorous framework specifically engineered for Landsat 9 thermal band data.

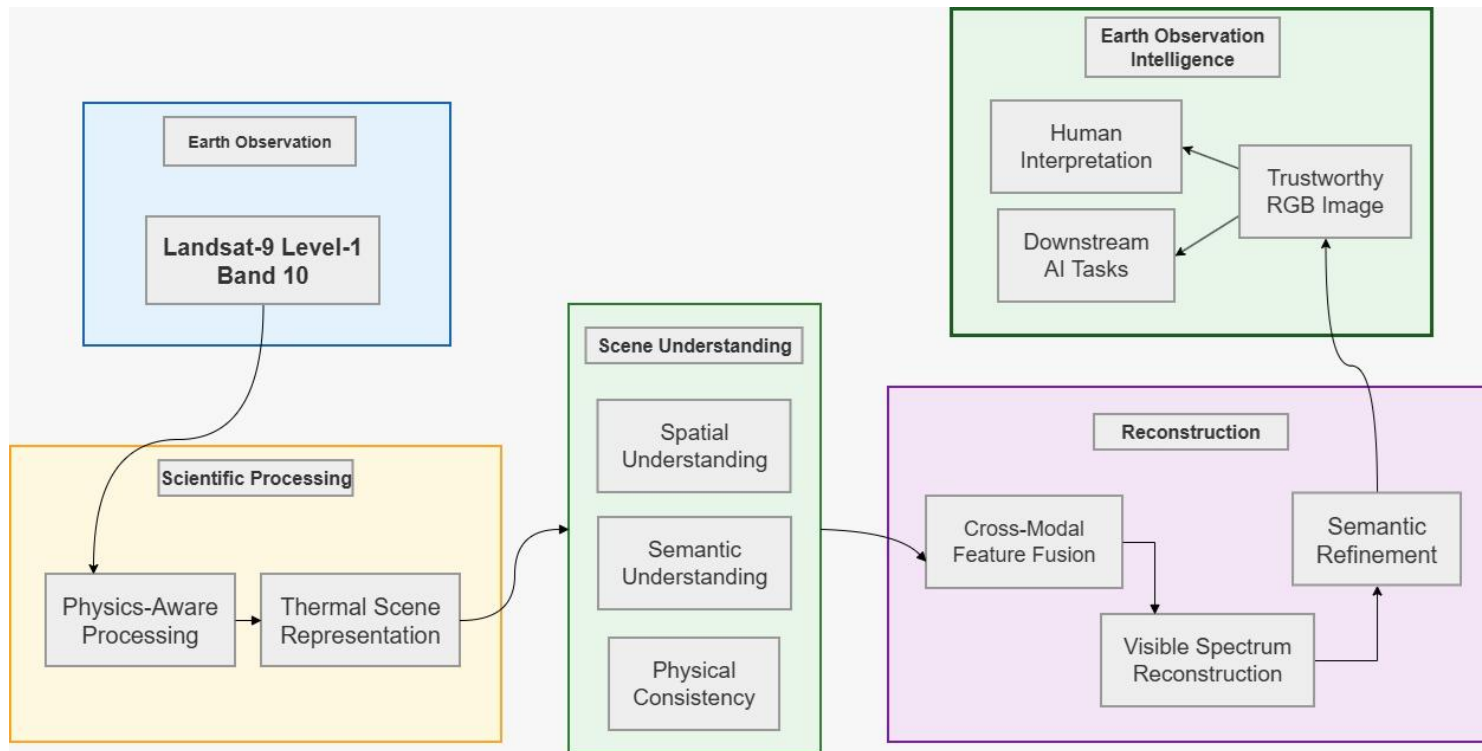
Key USPs

- Works on raw sensor data, not rendered images
- Semantic masks condition the generator, preventing water from being colorized brown, forests from being colorized grey
- Attention U-Net focuses learning on thermally faint but semantically critical boundaries
- Composite 4-term loss ensures the model optimizes for realism, structure, texture, and semantic correctness simultaneously
- Full geospatial output (GeoTIFF with CRS + transform preserved)

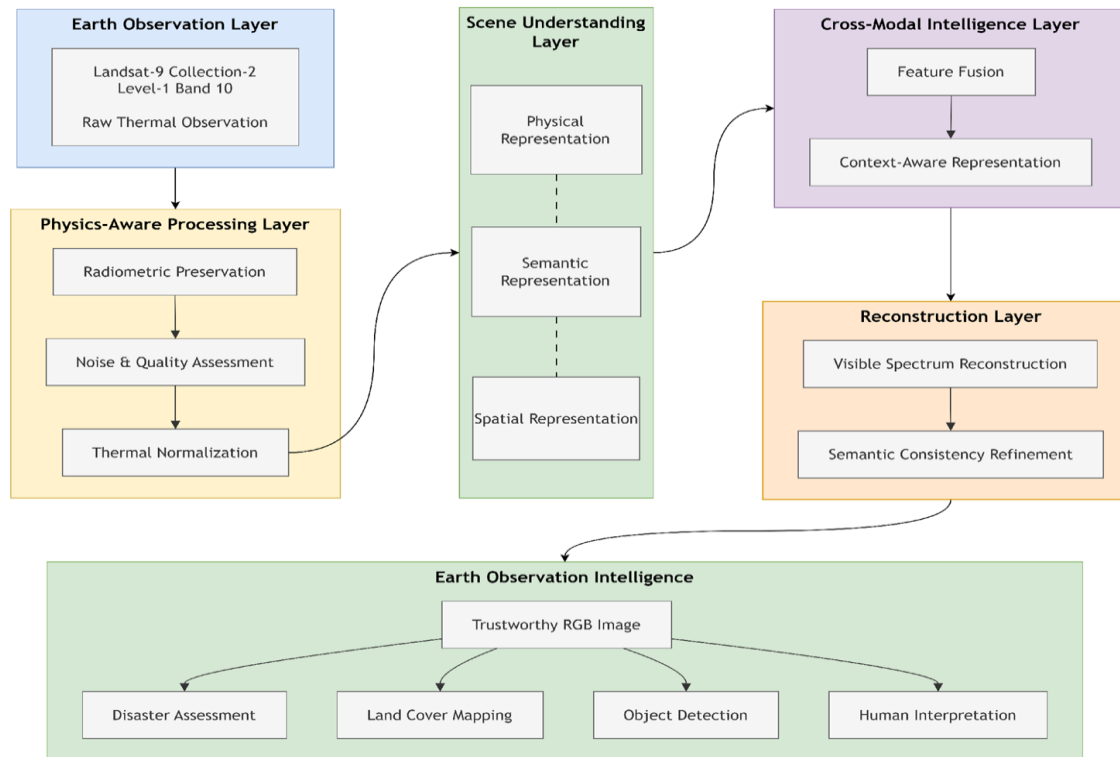
List of features offered by the solution

- 1. Physics-Aware Preprocessing:** Raw GeoTIFF Band 10 $\rightarrow$ DN to Brightness Temperature (Kelvin) conversion, QA-band cloud masking, radiometric normalization, 256x256 patch extraction.
- 2. Super-Resolution (Real-ESRGAN):** Fine-tuned Real-ESRGAN sharpens thermal patches from 100m to 30m-equivalent resolution, recovering faint edges like roads, buildings, and water boundaries.
- 3. Semantic Land-Cover Guidance (SegFormer):** Pretrained SegFormer generates per-pixel land-cover masks (Water | Vegetation | Urban | Bare Soil) used to condition the colorization model, ensuring semantically correct color mapping.
- 4. Semantic-Guided Colorization (Pix2Pix + Attention U-Net):** Attention U-Net generator takes a 5-channel input (SR thermal + semantic mask) and outputs a 3-channel RGB image. PatchGAN discriminator ensures local texture realism. Loss = Adversarial + L1 + Perceptual (VGG) + Semantic Constraint.
- 5. Post-Processing & Tile Stitching:** Histogram matching, Gaussian overlap blending, artifact suppression, and full GeoTIFF reconstruction with preserved CRS and spatial metadata.
- 6. Evaluation Suite:** PSNR | SSIM | LPIPS | FID | Inference Time per tile.
- 7. Interactive Demo Dashboard:** Before/After IR-to-RGB viewer with semantic mask overlay and per-tile metrics (Gradio/Streamlit).

Process flow diagram



Architecture diagram of the proposed solution



Technologies to be used in the solution:

- **Deep Learning Frameworks**

PyTorch & TorchVision (Core architecture, model training, and perceptual feature extraction)

- **Geospatial Data Engineering**

Rasterio & GDAL (Processing raw Landsat-9 GeoTIFFs, physical DN extraction, and CRS preservation)

- **Foundation Models**

Real-ESRGAN (Super-Resolution enhancement)

SegFormer / Hugging Face (Semantic land-cover masking)

- **Image Processing**

OpenCV & NumPy (High-performance array manipulation and patch stitching)

- **Evaluation Metrics**

Scikit-image (PSNR, SSIM), LPIPS (Perceptual similarity), PyTorch-FID (Realism score)

- **Deployment & MLOps**

Gradio / Streamlit (Interactive presentation dashboard)

Weights & Biases (W&B) (Real-time experiment tracking)

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THANK YOU